



ARISTOTLE UNIVERSITY OF THESSALONIKI

Faculty of Sciences

School of Physics

**Interpreting Neural Network Predictions
for Analog Circuit Performance
using SHAP Values**

Undergraduate Thesis

by

Michalis Deves

Supervisor Thomas Noulis

September 6, 2026

– This page has been intentionally left blank –

Acknowledgements

I would like to warmly thank my thesis supervisor, Dr. Thomas Noulis, for his guidance, trust, and substantial feedback during all stages of this study. His contribution was crucial in shaping the topic and the successful completion of this work.

Special thanks are owed to Eleni Papageorgiou for the significant assistance she provided, the fruitful discussions, and her insightful suggestions during the research and writing process.

Finally, I express my sincere gratitude to my family and friends for their unwavering support, patience, and encouragement throughout my studies.

Abstract

This thesis examines the prediction of key performance metrics of an operational amplifier via an artificial neural network and the subsequent interpretation of the predictions using Explainable AI methods. A fully connected neural network is implemented in PyTorch, trained on a dataset with five input parameters and eight output metrics. Preprocessing includes feature normalization and train/test split, while training is performed using Adam optimizer and MSE loss function. Evaluation on the test set shows good fit ($R^2 \approx 0.91$, $MSE = 0.0078$, $RMSE = 0.0883$), confirming the model's ability to capture nonlinear relationships between design parameters and performance metrics.

For interpretation, SHAP (Kernel Explainer) values are applied, which quantify the contribution of each input to the final prediction. The results reveal distinct roles of the parameters: C_c is critical for metrics such as Unity Gain and Slew Rate, W_{bp} dominates $ICMR_{max}$ and maximum output voltage, while W_{diff} and W_{nb} primarily affect $ICMR_{min}$ and minimum output voltage. The combined use of prediction and interpretability offers clear advantages over relying solely on time-consuming SPICE simulations, especially in rapid design optimization scenarios.

Overall, the approach demonstrates that neural networks, enriched with interpretation methods like SHAP values, constitute an efficient and transparent tool for the analysis and design of analog integrated circuits.

Keywords: artificial neural networks, explainable artificial intelligence, SHAP, operational amplifiers, analog integrated circuits, PyTorch.

Table of Contents

Acknowledgements	3
Abstract	4
1 Introduction	7
2 Theoretical Background	9
2.1 Neural Networks	9
2.2 Analog Integrated Circuits	10
2.3 AI Interpretation and SHAP	11
3 Methodology	14
3.1 Data	14
3.2 Model Architecture and Training	15
3.2.1 Neural Network Architecture	15
3.2.2 Cost Function and Optimization	16
3.2.3 Training Process	16
3.2.4 Model Evaluation	17
3.2.5 Interpretability with SHAP	18
3.3 Calculation and Analysis of SHAP Values	18
4 Results and Analysis	20
5 Conclusions	32
6 Bibliography	34

List of Figures	37
------------------------	-----------

Chapter 1

Introduction

The progress of artificial intelligence in recent years has enabled the application of modern machine learning techniques to a wide spectrum of scientific and technological fields. Among these, neural networks stand out for their ability to learn complex nonlinear relationships from data and achieve high accuracy in prediction problems.

An interesting application area for neural networks is that of analog integrated circuits. Numerous recent works have shown that ANNs can function as fast surrogate models[4] for critical subcircuits, accelerate device sizing[2] and predict performance (e.g., in operational amplifiers[1]) with high accuracy, significantly reducing simulation cost[1]. Specifically, ANN-based macro-models for op-amps[1] have been presented, offering multiple times the speed of corresponding SPICE simulations without substantial loss of accuracy, as well as ML-driven optimization frameworks[3] that learn from simulation data and effectively guide the design. In this context, the present work explores the theoretical and practical connection between neural networks and operational amplifiers, aiming to predict the performance of an operational amplifier through a suitably trained neural network, and interpret the predictions using SHAP values.

Although neural networks exhibit high performance, the interpretability of their decisions remains limited, which constitutes a significant obstacle in critical applications. To address this issue, the field of Explainable AI (XAI) has been developed, which includes methods that help understand the operation of a model and highlight the inputs that most influence its output.

To tackle this challenge, this thesis utilizes SHAP (SHapley Additive exPlanations) values, which are based on game theory and allow for the quantitative assessment of each input's contribution to the final prediction of the model.

The combination of modeling via neural networks and interpretation through SHAP values is applied to data related to the operation of an operational amplifier. Through this study, an attempt is made to highlight the critical factors affecting the

circuit's performance, as well as the extent to which each factor influences the final result.

Chapter 2

Theoretical Background

2.1 Neural Networks

Neural networks constitute a powerful class of computational models in the field of machine learning, aimed at approximating complex, often nonlinear, relationships between inputs and outputs. They are based on an architecture inspired by the functioning of biological neurons, where the model consists of successive layers of artificial neurons: the input layer, one or more hidden layers, and the output layer. Each artificial neuron performs a linear transformation of its inputs, followed by the application of a nonlinear activation function, such as ReLU, sigmoid, or tanh, with the purpose of increasing the network's expressive capacity[15].

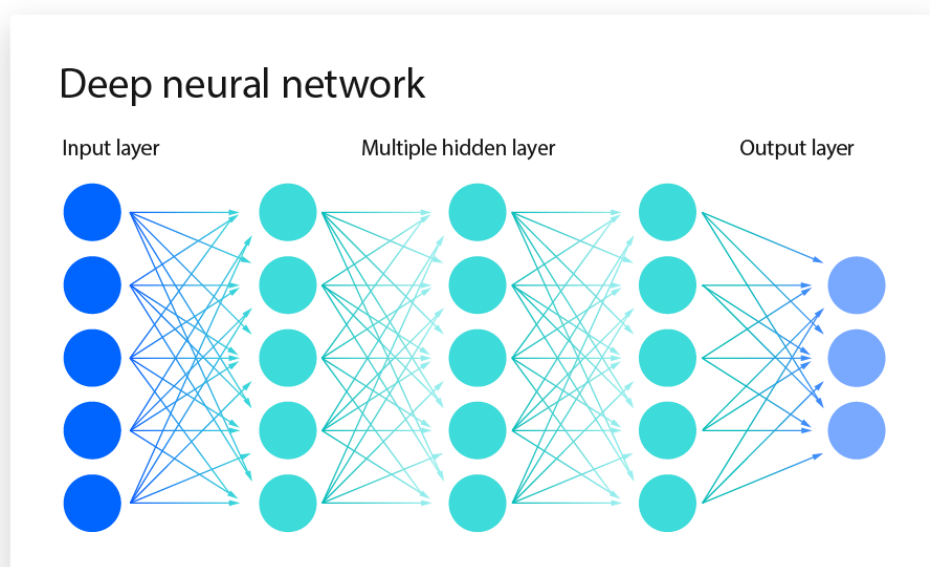


Figure 2.1: Architecture of a typical neural network [14].

The training process of neural networks involves the optimization of the model's

parameters, weights and biases, through appropriate algorithms such as gradient descent and backpropagation. Training aims to minimize a predefined cost function, which measures the difference between the model's predictions and the actual values. This approach makes neural networks particularly capable of solving high-complexity problems for which traditional techniques present limitations.

Due to their flexibility and ability for automatic feature extraction, neural networks have found extensive application in a wide spectrum of scientific and technological fields, such as pattern recognition, natural language processing, image analysis, and time series prediction. The continuous progress in the field of computational power, especially through the use of GPUs, has enabled the development of deeper and more efficient architectures, significantly enhancing the accuracy and applicability of neural networks to real-world problems.

2.2 Analog Integrated Circuits

Analog integrated circuits form a fundamental subset of integrated circuits, intended for processing analog signals, i.e., signals that vary continuously in time and take continuous values within their range. In contrast to digital circuits, which handle discrete logical states, analog circuits operate with voltages or currents representing real physical quantities. Main structural units of analog circuits include amplifiers, filters, oscillators, and modulators, with the operational amplifier (OpAmp) being one of the most important and widely used building blocks.

The design process of integrated circuits typically follows successive stages. It begins with specification definition, where the circuit's requirements in terms of operation and performance are determined. Subsequently, the appropriate topology is selected, according to the desired performance and the constraints of the implementation technology. This is followed by the circuit sizing stage, during which the parameter values of the transistors and passive elements are determined so that the circuit meets the specifications. The final stage is physical implementation at the layout level, where the geometry of the elements in the integrated circuit is designed, taking into account parasitic capacitance and resistance effects. Among the above stages, the present work focuses on circuit sizing, which is directly connected to the prediction and analysis of analog integrated circuit performance.

The design of analog integrated circuits requires high precision, as even small deviations in transistor parameters or passive element values can significantly affect the overall circuit behavior. Furthermore, factors such as power consumption, thermal noise, linearity, and bandwidth constitute critical points during the design process. For estimating the response of such circuits, analytical simulation tools are widely used, with SPICE (Simulation Program with Integrated Circuit Empha-

sis) [16] being the established standard. SPICE implements numerical methods for solving nonlinear differential equations that arise from the circuit, and allows the calculation of transient response (i.e., the time-domain behavior of the circuit when input signals are applied), DC response (operating point and DC sweeps), and AC response (frequency behavior, gain, and phase). The accuracy of the simulation depends greatly on the element models used; for MOSFETs, for example, the BSIM (Berkeley Short-channel IGFET Model) standards [17] are extensively applied, which describe microscale phenomena in detail, such as diffusion capacitances, parasitic resistances, and short-channel effects. Although SPICE-type simulations offer high accuracy, they are often computationally expensive, particularly when the circuit contains a large number of transistors or extensive parametric variation analysis is required. In such cases, simulation may take from minutes to hours for a single operating scenario, making its integration into rapid optimization processes difficult.

For this reason, in recent years, the utilization of artificial intelligence techniques—such as neural networks—has been explored as an alternative or complementary method for predicting the behavior of analog circuits, allowing faster performance estimation and better understanding of the factors influencing their response.

2.3 AI Interpretation and SHAP

As artificial intelligence models—and particularly deep neural networks—are increasingly applied in high-stakes domains such as medical diagnosis [5], autonomous systems [6], and financial data analysis [7], the need for interpretable and transparent decisions becomes evident. Despite their high performance, these models often operate as "black boxes," making it difficult to understand their internal mechanisms and the factors leading to a specific prediction or recommendation. The field of Explainable Artificial Intelligence (XAI) attempts to bridge this gap by developing methods that make the behavior of algorithms understandable, both at a global and local level.

One of the most established and theoretically grounded techniques within XAI is the SHAP (SHapley Additive exPlanations) method. This approach is based on game theory, where each input feature is treated as a "player" in a cooperative game and the goal is to fairly distribute the "payoff" (i.e., the model's prediction) among all players. Specifically, the Shapley value of a feature i is defined as the average marginal contribution when added to all possible combinations of feature subsets $S \subseteq N \setminus i$:

$$\phi_i(f) = \sum_{S \subseteq N \setminus \{i\}} \frac{|S|! (|N| - |S| - 1)!}{|N|!} [f(S \cup \{i\}) - f(S)],$$

where N is the set of all features and $f(S)$ is the model's output when only the

features of subset S are available.

The above formula ensures certain fundamental properties, such as efficiency (the sum of all ϕ_i equals the actual model prediction), symmetry (features with equal contribution receive equal values), absence (features that do not affect the output have zero value), and linearity (if the model is a linear combination, the Shapley values sum up) [18, 19].

SHAP values are calculated based on the average marginal contribution of each feature across all possible input combinations and provide a consistent, mathematically justified assessment of variable importance. The application of the SHAP method allows understanding of how the model makes its decisions, enhances transparency and user trust, and contributes to identifying potential biases, anomalies, or regions of the input space where the model behaves unexpectedly. In applications requiring high reliability and explainability, such as electronic circuit analysis, the integration of techniques like SHAP offers significant advantages both in terms of understanding and in design optimization processes. However, Shapley values are only as reliable as the data on which they are based; if the training data does not include sufficient observations to capture the effect of a feature, its contribution may be underestimated or distorted. Furthermore, the calculation process of Shapley values, which often relies on sampling from all possible input combinations, is limited by the distribution and density of the data in the input space. Thus, regions of the input space that are not adequately covered lead to less accurate estimates, while strong dependencies or rare conditions may not be fully revealed. Overall, the reliability of Shapley values is closely linked to the quality, variety, and density of the training data.

A particularly widespread implementation of the SHAP method is the Kernel Explainer [20, 21]. This is an algorithm that allows approximate calculation of Shapley values for any model, regardless of its internal structure, treating it as a "black box." The Kernel Explainer is based on a kernel weighting technique, where different feature combinations are sampled and then the contribution of each feature to the prediction is estimated using a weighted linear regression.

Mathematically, the Kernel Explainer approximates the prediction of the original model $f(x)$ through a linear explanatory model of the form:

$$f(x) \approx g(z') = \phi_0 + \sum_{j=1}^M \phi_j z'_j,$$

where z'_j are binary variables indicating the presence or absence of feature j , while the coefficients ϕ_j represent the corresponding SHAP values. The value ϕ_0 corresponds to the average prediction of the model when no input is available.

The weighting of different feature subsets is achieved through a kernel function, which assigns a weight to each subset:

$$\pi_z = \frac{(M - 1)}{\binom{M}{|z'|} |z'| (M - |z'|)},$$

where M is the total number of features and $|z'|$ is the size of the specific subset. In this way, subsets that include more features and better approximate the complete set of inputs receive greater weight, ensuring that the regression remains faithful to the behavior of the original model.

The term kernel refers to a mathematical weighting function that determines how important each feature subset is considered during the calculation. In practice, subsets that more closely resemble the complete set of inputs receive greater weight, while those with few features receive less. In this way, the regression aligns with the actual behavior of the model, leading to more reliable estimation of Shapley values.

This method has the advantage that it can be applied to any machine learning algorithm, without requiring knowledge or modification of its internal mechanisms. However, its computational cost depends heavily on the number of features, as the number of necessary samplings increases exponentially. Despite this limitation, the Kernel Explainer is widely used in practice due to the flexibility and theoretical validity it provides, constituting a fundamental tool for understanding and explaining complex models on real data.

Beyond the SHAP method, other widely used interpretability techniques have been developed in the field of explainable artificial intelligence. For example, the LIME (Local Interpretable Model-agnostic Explanations) method [8] provides local linear approximations around each prediction of the model, allowing understanding of the features that influence a specific decision. Another category includes methods based on sensitivity functions and derivatives, such as Integrated Gradients [9], which distributes the prediction to the input features through the integration of gradients along a path from a reference point. Furthermore, visualization techniques such as Grad-CAM [10] have particular application in convolutional neural networks, as they reveal which regions of the image contribute most to the prediction.

These different approaches highlight the wide range of tools available in the field of XAI, each with advantages and limitations depending on the type of model and application domain. In the present work, the use of SHAP is selected because it provides a theoretically grounded and consistent distribution of input contributions to outputs, is independent of the model type, and can be effectively applied to multivariate regression problems.

Chapter 3

Methodology

The methodology followed in this work aims at the development, training, and interpretation of an Artificial Neural Network (ANN) model for predicting key performance metrics of an operational amplifier (Figure 3.1). The process includes four main stages: (a) data collection and preprocessing, (b) model design and training, (c) performance evaluation, and (d) calculation and analysis of SHAP values for interpretation of the predictions.

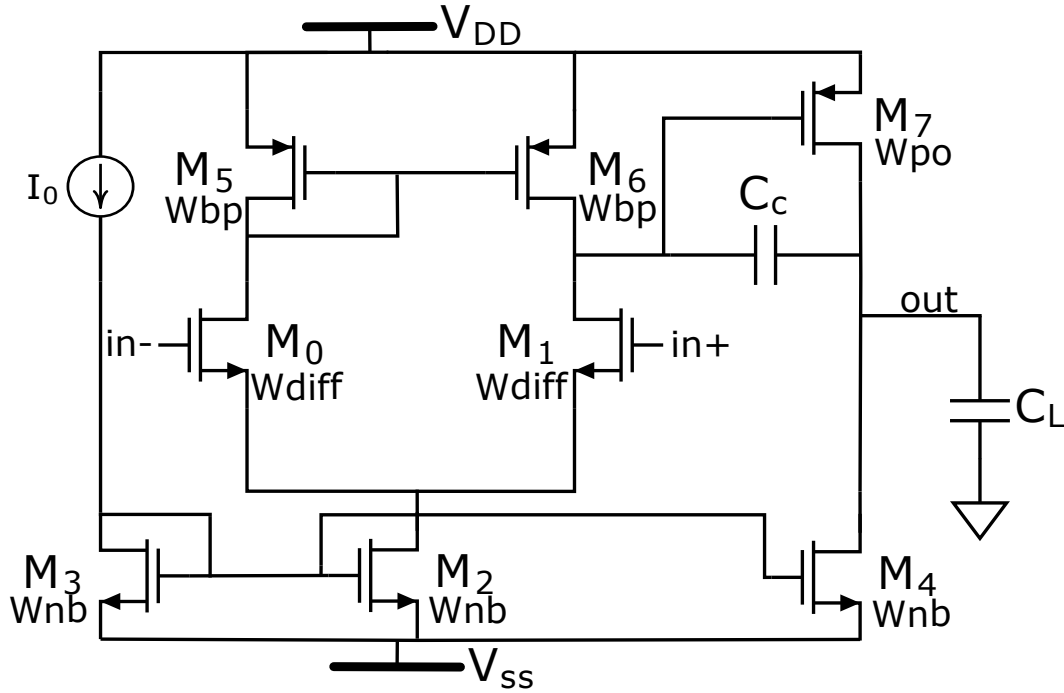


Figure 3.1: Operational Amplifier Circuit.

3.1 Data

Table 3.1 presents in detail the circuit parameters and performance metrics.

Table 3.1: Circuit Parameters and Performance Metrics

Category	Abbreviation	Full Name / Description
Parameters	Cc	Coupling Capacitor
	Wbp	Biasing point transistor width
	Wdiff	Differential pair width
	Wnb	Negative bias / NMOS width
	Wpo	Positive output / PMOS width
Performance Metrics	PM	Phase Margin
	UGBW	Unity Gain Bandwidth
	G	Gain
	ICMR_max	Maximum Input Common-Mode Range
	ICMR_min	Minimum Input Common-Mode Range
	Vout_max	Maximum output voltage
	Vout_min	Minimum output voltage
	SR	Slew Rate

The initial data format was a CSV file, which was loaded using the pandas library [11].

Data processing included:

1. **Completeness and consistency check:** It was ensured that there were no missing or invalid values that could affect training.
2. **Feature normalization:** Standard Scaling from the scikit-learn library [22] was applied (zero mean and unit standard deviation) to both inputs and outputs. Normalization reduces the probability of numerical instability during optimization and equalizes the scale of the features.
3. **Subset splitting:** The dataset was split into a **training set** (80%) and a **test set** (20%) using a fixed random seed for reproducibility.

3.2 Model Architecture and Training

The neural network architecture was designed to predict eight analog circuit performance metrics based on five input parameters. Implementation was carried out in the PyTorch environment [12] and is based on a fully connected feed-forward network.

3.2.1 Neural Network Architecture

The network consists of:

- A five-dimensional input layer,

- Two hidden layers of fully connected neurons with ten units each,
- A Rectified Linear Unit (ReLU) activation function after the first hidden layer and another ReLU after the second hidden layer,
- An eight-dimensional output layer, corresponding to the circuit's performance metrics.

The forward pass process is described as follows:

$$x \longrightarrow \text{Linear}(5, 10) \longrightarrow \text{ReLU} \longrightarrow \text{Linear}(10, 10) \longrightarrow \text{ReLU} \longrightarrow \text{Linear}(10, 8) \longrightarrow \hat{y}$$

Note: In this specific model, an activation function is also inserted between the two fully connected hidden layers. Thus, the network gains greater non-linearity and expressive capacity without exhibiting instability during training. This choice improved the model's generalization and led to satisfactory results in the experiments ($R^2 \approx 0.91$).

The use of the ReLU function helps avoid the vanishing gradients problem. This phenomenon occurs when activation functions with limited value ranges are used, such as sigmoid or tanh, resulting in the derivatives propagated backwards during the backpropagation algorithm becoming extremely small. As the network depth increases, these very small values tend to zero, causing training to slow down or "freeze," as the weights of the first layers are not updated sufficiently.

ReLU, in contrast, maintains a non-zero derivative for all positive inputs, allowing gradually greater information flow backwards and facilitating optimization. Thus, convergence is achieved faster and the network can learn complex patterns more effectively.

3.2.2 Cost Function and Optimization

For model training, the Mean Squared Error (MSE) cost function is used:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N (y_i - \hat{y}_i)^2$$

Parameter optimization is performed using the Adam algorithm [13], which was selected due to its efficiency in convergence on complex problems. The learning rate was set to 0.001.

3.2.3 Training Process

The network was trained for 3000 epochs, with weight updates via backpropagation. For training, 837 samples were used (80% for training and 20% for testing),

where each sample corresponds to a combination of five input parameters and eight performance metrics. The error progression is monitored by printing the value of the cost function (Mean Squared Error loss) every 100 epochs. This is implemented in the training loop with a control condition that activates the print command and displays the current error on the screen. Thus, the user is given the ability to monitor the convergence progress in real time and identify possible instabilities or overfitting problems.

During training, the training loss progression is recorded at each epoch, while the corresponding convergence curve is created simultaneously. Figure 3.2 shows the variation of the cost function (MSE) value with respect to the number of epochs. A rapid decrease in error is observed in the first hundreds of epochs and gradual stabilization as the model converges. At the end, the model is saved to a `performance_model.pth` file, along with the input and output normalizers, so that the results can be reproduced and evaluated on future data.

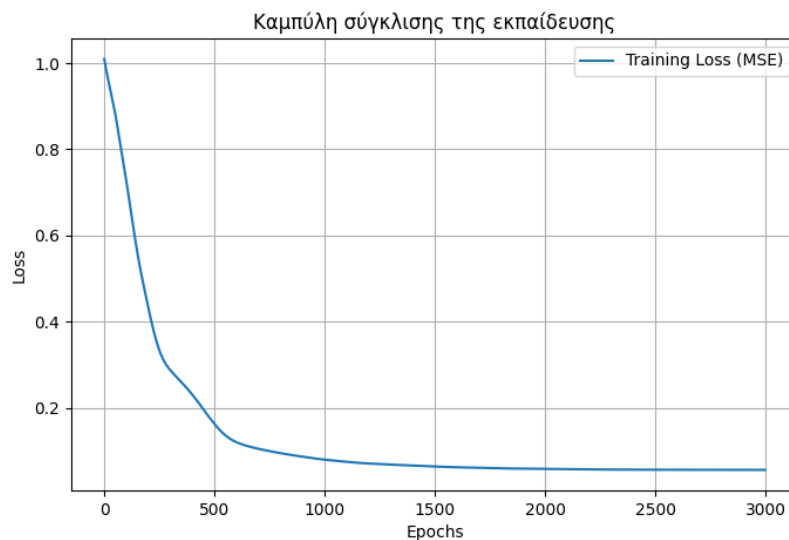


Figure 3.2: Convergence curve of the cost function (MSE) during the training of the neural network for 3000 epochs.

3.2.4 Model Evaluation

The model's performance on the test set is evaluated using three metrics:

- MSE (Mean Squared Error),
- RMSE (Root Mean Squared Error),
- Coefficient of Determination R^2 .

Additionally, scatter plots are created between the actual and predicted values, providing a visual assessment of the model's performance.

3.2.5 Interpretability with SHAP

For interpretability analysis, the SHAP method was used. For each of the eight model outputs, the KernelExplainer was applied, which estimates the effect of each input on the formation of the final prediction.

The results are presented in the form of summary plots, generated through the SHAP library [19], which capture the relative importance of the features, as well as with tables including the mean and standard deviation of the SHAP values per feature. This analysis provides substantial information about the contribution of inputs to outputs, making it possible to understand the relationship between design parameters and the performance of analog circuits.

3.3 Calculation and Analysis of SHAP Values

The interpretation of predictions was performed through the SHAP method, which is based on game theory and provides a quantitative estimate of each feature's contribution to the final prediction. The method assigns a Shapley value to each input, representing the average marginal contribution of the input across all possible feature combinations.

For this work, the KernelExplainer was selected because:

- It is independent of the model type and architecture.
- It provides consistent results in multivariate regression problems.
- It can be applied separately for each output, which is necessary in this specific study.

The process was implemented as follows:

1. Selection of a **reference set** (background data) from 50 samples of the training set, in order to calculate the model's baseline predictions.
2. Selection of 100 samples from the test set for analysis.
3. For each of the 8 outputs, SHAP values were calculated via the KernelExplainer.
4. Creation of summary plots that visualize the importance and distribution of SHAP values for each feature.
5. Calculation of statistical indicators (mean and standard deviation) of the SHAP values per feature, for the purpose of quantitative ranking of their importance.

The application of SHAP values allows the identification of critical inputs that most affect the performance of the operational amplifier and enhances the transparency and reliability of the proposed model.

Chapter 4

Results and Analysis

This section presents the results of the experimental procedure. Initially, the model's performance on the test set is evaluated using appropriate metrics and graphs. Subsequently, the interpretability of the predictions is examined through the SHAP method, in order to highlight the contribution of input parameters to the performance metrics of operational amplifiers.

After completing the training, the model was evaluated on the test set, achieving the following results:

- Test Loss (MSE): 0.0078
- Root Mean Squared Error (RMSE): 0.0883
- **Coefficient of determination (R^2): 0.9125**

These results show that the model has good predictive capability for the performance metrics of operational amplifiers, with small error and high linear correlation between actual and predicted values.

Figure 4.1 shows the scatter plot of actual versus predicted values for the test set. The red dashed line represents the ideal correspondence ($y = x$), confirming the model's good performance.

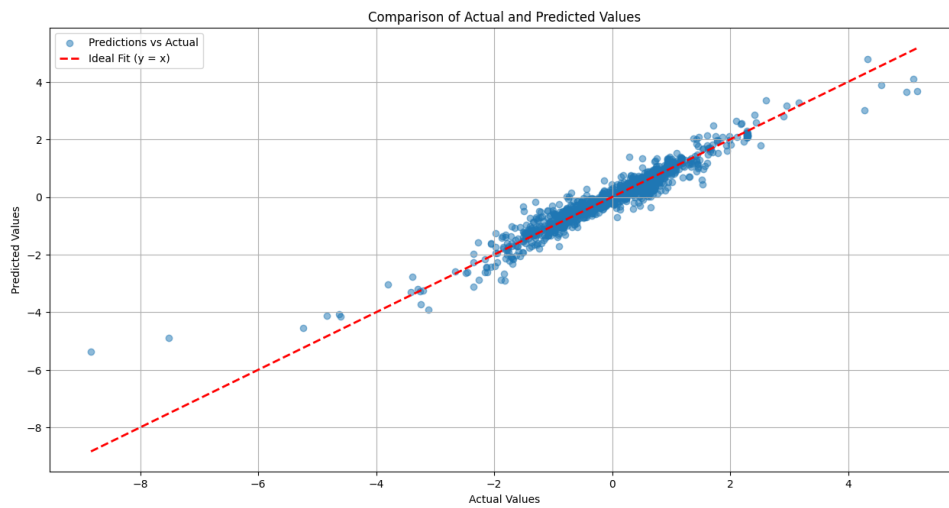


Figure 4.1: Comparison of actual and predicted performance metric values on the test set.

For each of the 8 performance metrics, SHAP values were calculated and visualized in the corresponding summary plots. From these diagrams (presented below), it is evident that certain input features have greater influence on the prediction of specific outputs. Subsequently, the interpretation of the SHAP plots is presented, aiming to clarify how individual features affect the model's outputs. The application of SHAP values improves the model's transparency and helps understand the relationships between inputs and outputs, which is critical for the optimization and design of operational amplifiers.

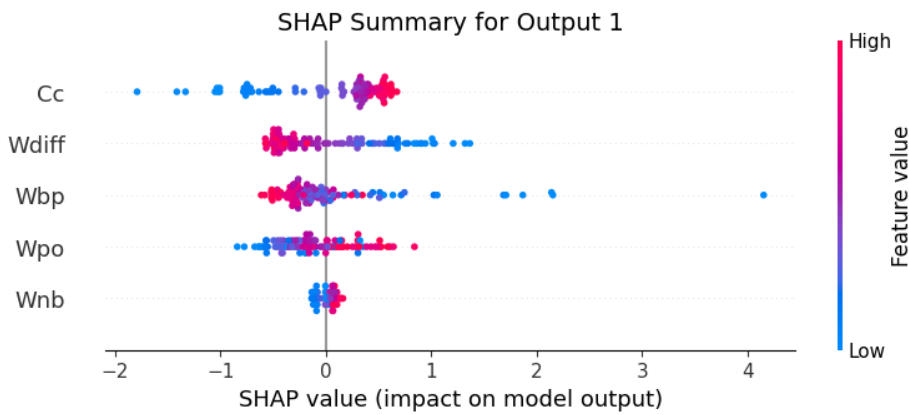


Figure 4.2: SHAP summary plot for the model output.

In Figure 4.2, the vertical axis (Y) displays the input features of the model, sorted in descending order of importance as derived from the mean absolute value of the SHAP values. In this diagram, parameter Cc shows the greatest influence on the model's predictions, while parameter Wnb shows the least.

The horizontal axis (X) presents the SHAP values, which express the direction and magnitude of each feature's effect on the model's expected prediction value and not on the actual target value (label) used during training. Positive values indicate that the feature contributes to an increase in the prediction, while negative values indicate a decrease. It is worth noting that these values are in a normalized scale (e.g., in the range approximately $[-3, 3]$), since the model was trained with normalized outputs. Consequently, the actual effect in physical units (e.g., output voltage in Volt) is obtained through inverse normalization, in order to accurately express the magnitude of each parameter's contribution to the circuit output. The tables with the normalized values follow:

Table 4.1: Normalization limits (Parameters). For each parameter: μ (mean), σ (standard deviation), min (minimum value in sample), max (maximum value in sample) in *physical units*, and min z , max z (corresponding limits in the normalized space $z = \frac{x-\mu}{\sigma}$).

Parameter	μ	σ	min	max	min z	max z
Cc	10.8529	4.58791	1	21	-2.15	2.21
Wbp	12.2895	4.22936	1	21	-2.67	2.06
Wdiff	2.37703	0.968894	0.3	4.3	-2.14	1.98
Wnb	7.33373	5.10487	1	21	-1.24	2.68
Wpo	27.3648	8.40894	10	41	-2.07	1.62

Table 4.2: Normalization limits (Performance Metrics). For each metric: μ (mean), σ (standard deviation), min (minimum value in sample), max (maximum value in sample) in *physical units*, and min z , max z (corresponding limits in the normalized space $z = \frac{x-\mu}{\sigma}$).

Metric	μ	σ	min	max	min z	max z
PM	57.5625	10.7359	24.4329	110.304	-3.09	4.91
UG	2.40996e+06	1.21153e+06	117490	9.12011e+06	-1.89	5.54
G	50.3403	13.0504	6.99648	62.3874	-3.32	0.923
ICMR_max	0.856777	0.0458681	0.509314	0.918	-7.58	1.33
ICMR_min	-0.288238	0.0543581	-0.397554	-0.157699	-2.01	2.4
Vout_max	0.791927	0.0206236	0.609689	0.817718	-8.84	1.25
Vout_min	-0.775584	0.0484074	-0.82882	-0.664045	-1.1	2.3
SR	3.25021	2.0628	0.996505	26.498	-1.09	11.3

The color of the points corresponds to the normalized value of the feature: blue represents low values, while red represents high values. The color scaling allows simultaneous evaluation of both the importance and the actual value of each feature, facilitating the interpretation of the model’s behavior. The quantification of each feature’s importance was performed through the mean and standard deviation of the absolute SHAP values, as presented in the following tables. These values provide a concise but substantial assessment of the relative contribution of parameters to the model’s predictions, since the mean $\text{mean}(|\phi|)$ expresses the overall importance, while the standard deviation $\text{std}(|\phi|)$ captures the dispersion of the effect across different samples.

Subsequently, the SHAP summary plots for each of the 8 performance metrics are presented separately:

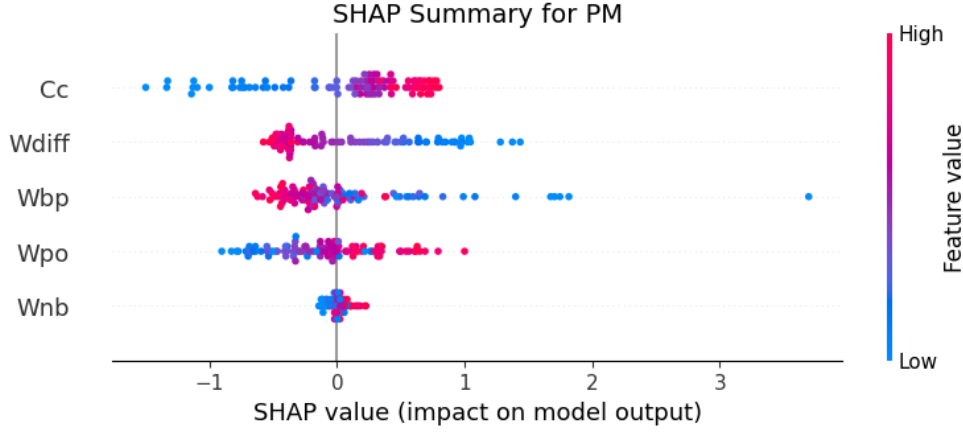


Figure 4.3: Phase Margin

Table 4.3: Mean values and standard deviations of SHAP for Output 1 Phase Margin

Feature	Mean SHAP Value	Standard Deviation
Cc	0.4696	0.3411
Wbp	0.3909	0.5488
Wdiff	0.4321	0.3255
Wnb	0.0481	0.0348
Wpo	0.3485	0.1882

For Output 1 Phase Margin, it is observed that parameters Cc (0.47 ± 0.34), Wbp (0.39 ± 0.55) and Wdiff (0.43 ± 0.33) show similar levels of significance, indicating that they equally affect the prediction. Parameter Wpo (0.35 ± 0.19) has a smaller but calculable contribution, while Wnb shows the minimum effect (0.05 ± 0.03). The scatter plot of SHAP values shows a relatively symmetric distribution, where positive values correspond to an increase in the prediction and negative to a decrease, without extreme deviations. This pattern agrees with operational amplifier theory, according to which the phase margin improves when the compensation capacitance C_c achieves sufficient pole splitting, while the geometric parameters of the stages (such as Wbp and Wdiff) affect the g_m and r_o and consequently the pole positions.

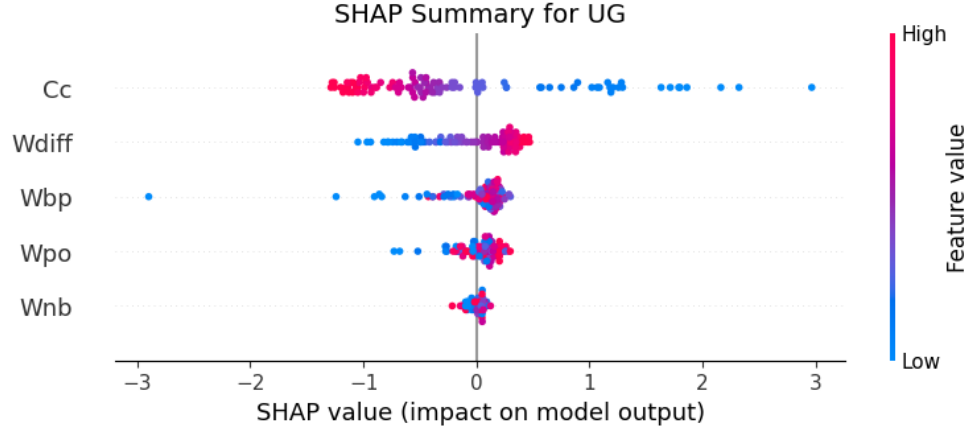


Figure 4.4: Unity Gain Bandwidth

Table 4.4: Mean values and standard deviations of SHAP for Output 2 Unity Gain Bandwidth

Feature	Mean SHAP Value	Standard Deviation
Cc	0.7926	0.5723
Wbp	0.2189	0.3124
Wdiff	0.3184	0.2463
Wnb	0.0285	0.0249
Wpo	0.1380	0.1089

For Output 2 Unity Gain Bandwidth, parameter C_c (0.79 ± 0.57) has dominant importance, directly affecting the output value: high C_c values are associated with a positive effect, while low values with a negative effect. Parameters W_{bp} and W_{diff} have secondary contribution, while W_{po} and W_{nb} play a minimal role. The diagram shows a wide distribution of points, with some extreme values, indicating that C_c can largely determine the output variation. As expected from theory, the unity gain bandwidth ($UGBW$) approximates g_{m1}/C_c , which explains why C_c appears dominant in the predictions. It is also worth noting that the sign of the SHAP values has a clear physical interpretation: high values of C_c (red points) correspond mainly to negative SHAP values, i.e., they contribute to a decrease in $UGBW$, while low values of C_c (blue points) lead to positive SHAP values, contributing to an increase in $UGBW$.

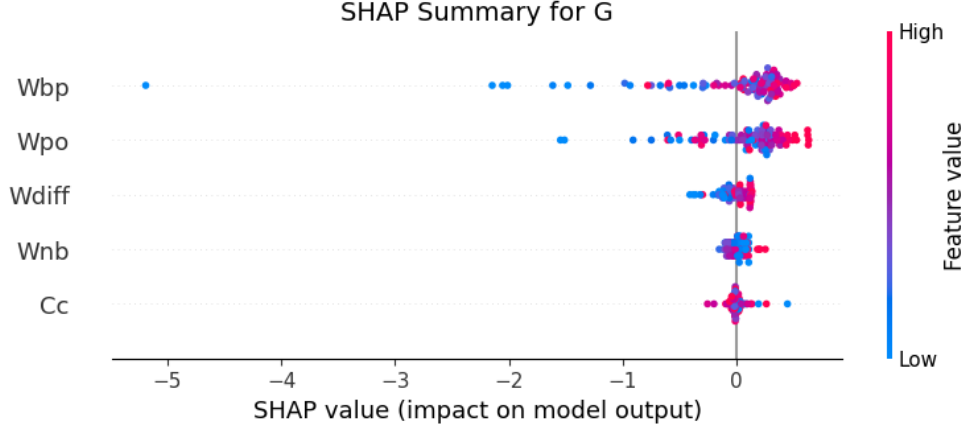


Figure 4.5: Gain

Table 4.5: Mean values and standard deviations of SHAP for Output 3 Gain

Feature	Mean SHAP Value	Standard Deviation
Cc	0.0282	0.0267
Wbp	0.4568	0.5250
Wdiff	0.0874	0.0799
Wnb	0.0660	0.0431
Wpo	0.3430	0.2518

For Output 3 Gain, parameter Wbp is the most significant (0.46 ± 0.53), while Wpo also has a significant contribution (0.34 ± 0.25). The remaining parameters have lesser importance, with Cc appearing almost negligible. The point distribution shows that low values of Wbp are associated with a strong negative effect, while high values shift the prediction towards the positive side, revealing a nonlinear relationship. This picture is consistent with theory, according to which the DC open-loop gain depends on the product of the g_m and r_o of the stages, therefore the geometric parameters Wbp and Wpo play a primary role, while C_c does not significantly affect the static gain. More specifically, the open-loop gain can be approximated by the relation $A_v \approx g_m \cdot r_o$, where g_m is the transconductance and r_o is the output resistance of the MOSFET. The transconductance is defined as

$$g_m = \frac{\partial I_D}{\partial V_{GS}} \approx \frac{2I_D}{V_{OV}},$$

where I_D is the **drain current** and $V_{OV} = V_{GS} - V_{TH}$ is the gate overdrive voltage. The output resistance is given by

$$r_o \approx \frac{1}{\hat{\lambda} I_D},$$

where $\hat{\lambda}$ is the **channel-length modulation parameter**. From the above relations, it becomes clear that the gain depends directly on the drain current and the geometric

dimensions W/L of the transistors, as these parameters determine both g_m and r_o . More specifically, for MOSFET in saturation region:

$$I_D = \frac{1}{2} \mu C_{ox} \frac{W}{L} (V_{OV})^2,$$

so, for constant V_{OV} , the drain current is proportional to the width W . Consequently, in the biasing stage, the current I_D increases linearly with W_{bp} , leading also to an increase in transconductance g_m , since $g_m \approx 2I_D/V_{OV}$. In the output stage, parameter W_{po} determines the current I_D . For constant V_{OV} , an increase in W_{po} increases I_D , which decreases r_o . Thus, W_{po} mainly affects r_o and through it the open-loop gain.

It is worth noting that if the output current I_D is kept constant via a current source, then r_o does not change with W_{po} . In this case, changing W_{po} decreases the overdrive voltage V_{OV} , thus increasing $g_m = 2I_D/V_{OV}$. That is, W_{po} can affect $A_v = g_m r_o$ either through r_o or through g_m , depending on the biasing conditions. Thus, it is explained why parameters W_{bp} and W_{po} have a decisive contribution in determining the Gain, while the compensation capacitance C_c does not significantly affect the static gain.

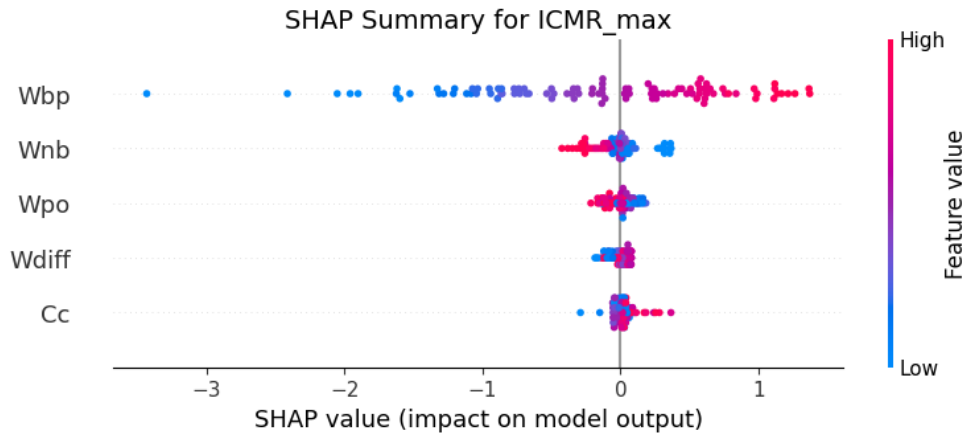


Figure 4.6: ICMR max

Table 4.6: Mean values and standard deviations of SHAP for Output 4 ICMR_max

Feature	Mean SHAP Value	Standard Deviation
Cc	0.0359	0.0753
Wbp	0.6865	0.7063
Wdiff	0.0610	0.0526
Wnb	0.0877	0.0563
Wpo	0.0209	0.0208

Output 4 ICMR-max is influenced almost exclusively by parameter W_{bp} (0.69 ± 0.71), which shows large dispersion, indicating that its different values strongly

alter the prediction. The remaining parameters have very low importance. The point distribution is asymmetric: low values of Wbp are associated with negative contribution, while high values with positive, demonstrating its determining role in shaping the output. This is directly connected to theory, as the upper common-mode limit is determined by the headroom (voltage margin) towards V_{DD} and the V_{OV} of the PMOS transistors in the upper branch of the circuit; therefore, the effect of Wbp is expected.

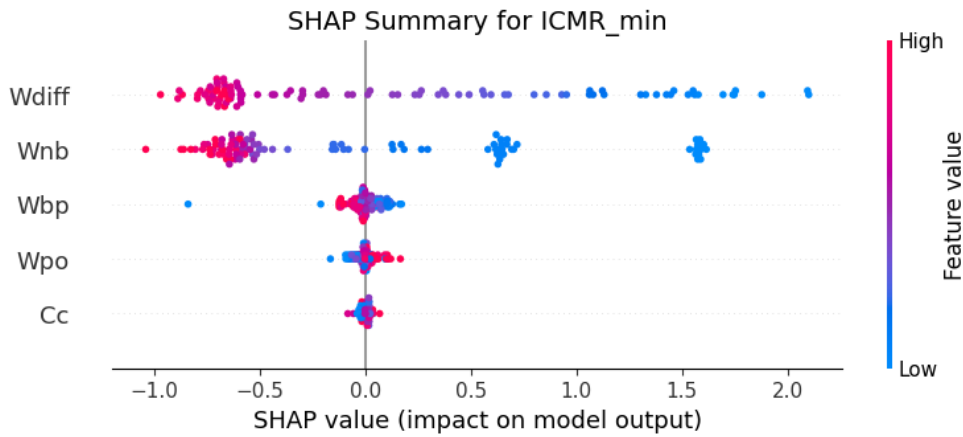


Figure 4.7: ICMR min

Table 4.7: Mean values and standard deviations of SHAP for Output 5 ICMR_min

Feature	Mean SHAP Value	Standard Deviation
Cc	0.0122	0.0099
Wbp	0.0271	0.0280
Wdiff	0.7302	0.5518
Wnb	0.7162	0.3857
Wpo	0.0184	0.0165

For Output 5 ICMR-min, the most important parameters are Wdiff (0.73 ± 0.55) and Wnb (0.72 ± 0.39), which largely determine the output. Wdiff shows a symmetric distribution around zero, with positive and negative values distributed evenly, while Wnb has a strong positive effect when it takes high values. The remaining parameters have negligible significance. This behavior reflects the theoretical prediction that the lower common-mode limit is constrained by the V_{OV} of the differential pair and the lower current source; hence, the geometric parameters Wdiff and Wnb determine ICMR_min.

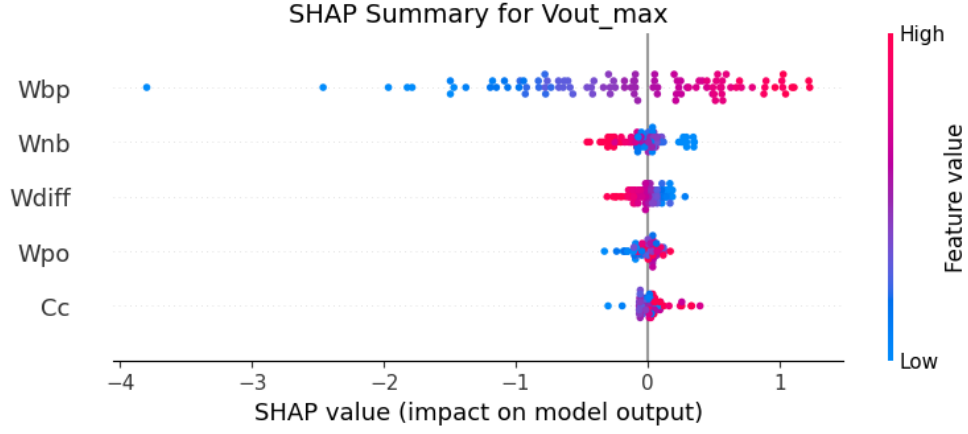


Figure 4.8: Vout max

Table 4.8: Mean values and standard deviations of SHAP for Output 6 Vout_max

Feature	Mean SHAP Value	Standard Deviation
Cc	0.0407	0.0894
Wbp	0.6157	0.7290
Wdiff	0.0958	0.1156
Wnb	0.0739	0.0518
Wpo	0.0842	0.0550

Output 6 Vout_max depends mainly on parameter Wbp (0.62 ± 0.73), which shows the greatest variation. The diagram shows that low Wbp values have a negative contribution, while high values lead to positive. The remaining parameters contribute minimally, with Cc showing a limited but clear positive trend. This finding agrees with theory, according to which the maximum swing towards V_{DD} is limited by the $V_{OV,p}$ of the output stage, which decreases as the W/L of Wbp increases. This is due to the fact that Wbp determines the circuit's biasing current, which in turn regulates the $V_{OV,p}$ of the output stage. Thus, through biasing, Wbp is directly connected to the maximum voltage swing towards V_{DD} .

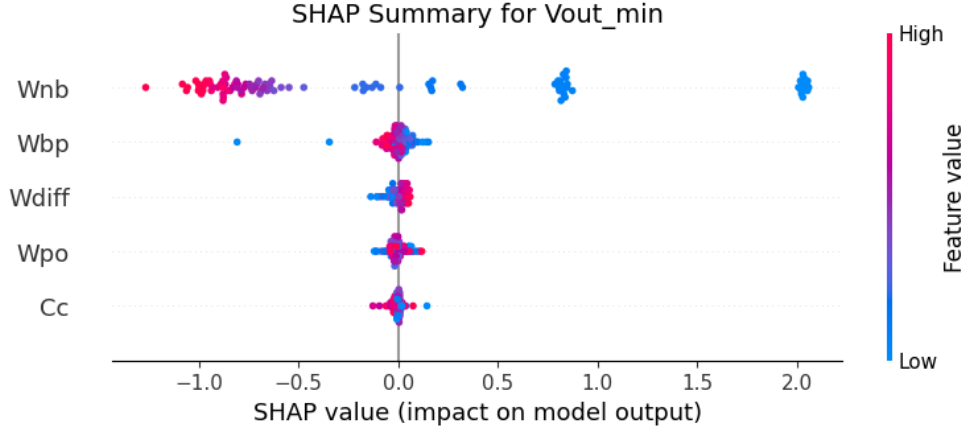


Figure 4.9: Vout min

Table 4.9: Mean values and standard deviations of SHAP for Output 7 Vout_min

Feature	Mean SHAP Value	Standard Deviation
Cc	0.0104	0.0089
Wbp	0.0090	0.0092
Wdiff	0.0479	0.0408
Wnb	0.9050	0.4800
Wpo	0.0110	0.0101

Output 7 Vout_min is completely dominated by parameter Wnb (0.91 ± 0.48), with the remaining features having practically zero significance. The distribution shows that high Wnb values lead to a significant positive effect, while low values to negative. The diagram clearly captures the one-dimensional dependence of the output on this parameter. This is in agreement with theory, as the minimum swing towards V_{SS} is determined by the $V_{OV,n}$ of the "lower" transistor; larger W/L leads to smaller $V_{OV,n}$ and thus lower V_{out_min} .

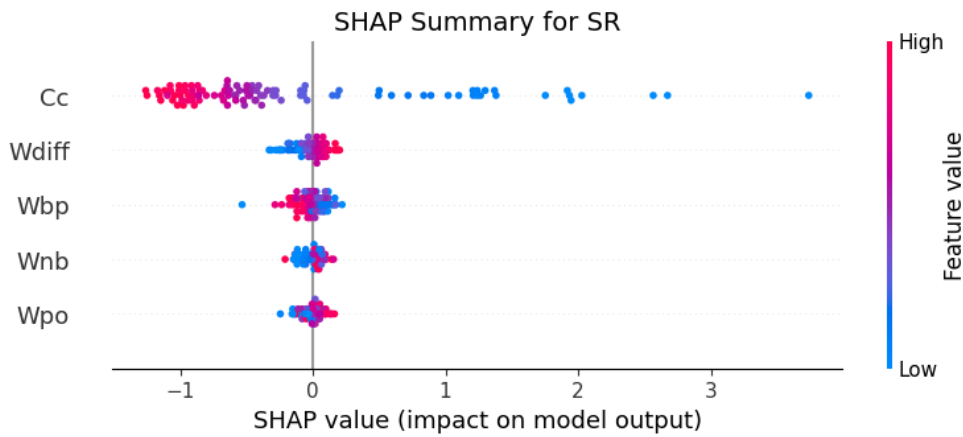


Figure 4.10: Slew Rate

Table 4.10: Mean values and standard deviations of SHAP for Output 8 Slew Rate

Feature	Mean SHAP Value	Standard Deviation
Cc	0.8437	0.6300
Wbp	0.0499	0.0613
Wdiff	0.1184	0.1021
Wnb	0.0519	0.0370
Wpo	0.0224	0.0283

For Output 8 Slew Rate, parameter Cc is decisive (0.84 ± 0.63), surpassing all others. The distribution shows that high Cc values are associated with positive contribution, while low with negative, capturing a clear dependence relationship. The remaining features have secondary or negligible significance. This dominance of C_c aligns with theory, as the slew rate is given by $SR \approx I_5/C_c$, so the compensation capacitance is the determining factor.

More specifically, the means and standard deviations of the SHAP values per feature (for each output) confirm the importance of functional parameters, such as transistor dimensions and biasing voltages, in the overall performance of the model.

Chapter 5

Conclusions

This thesis focused on the development and interpretation of an artificial neural network for predicting key performance metrics of an operational amplifier. The central objective was twofold: on one hand, the construction of a reliable prediction model that reduces dependence on time-consuming and computationally intensive SPICE-type simulations, and on the other hand, the in-depth understanding of the parameters that most affect the circuit's performance, through the use of explainable artificial intelligence methods.

The model training, implemented in the PyTorch environment, demonstrated its ability to adequately capture the nonlinear relationships between inputs and outputs. The high level of accuracy achieved on the test set (with $R^2 \approx 0.91$) confirms that neural networks can constitute an effective alternative approach to classical simulation methods.

The interpretability analysis through SHAP values allowed the quantitative assessment of each input's contribution to the model's final predictions, enhancing the transparency and reliability of the process. The results indicated that different performance metrics are affected to different degrees by individual parameters: the coupling capacitance (C_c) proved critical for metrics such as Unity Gain and Slew Rate, while parameter W_{bp} plays a primary role in metrics such as $ICMR_{max}$ and maximum output voltage. Simultaneously, parameters W_{diff} and W_{nb} emerged as critical factors for $ICMR_{min}$ and minimum output voltage. This differentiation underscores the importance of multifactorial analysis, since the performance of an operational amplifier does not depend unidimensionally on a single variable.

The contribution of the SHAP method is not limited merely to clarifying the model's predictions, but extends to the possibility of integrating these interpretations into design optimization processes. Compared to traditional simulations, the proposed approach offers significant advantages in terms of speed and interpretability, without sacrificing accuracy. Therefore, the use of neural networks in combination with explainable artificial intelligence methods can constitute a particularly useful

tool in the initial design and analysis of analog integrated circuits.

Overall, the findings of this work confirm that the utilization of modern machine learning techniques can substantially enhance the field of microelectronics. The capability of prediction with high accuracy, combined with the interpretability provided by the SHAP method, constitutes an innovative direction that can facilitate the design of more efficient and reliable circuits.

Future Extensions

Despite the encouraging results, the research leaves room for further investigation. A first step consists of expanding the dataset to cover a wider range of parameters and circuit architectures, thereby increasing the model's generalizability. In parallel, the study of more complex neural network architectures, such as convolutional or recurrent networks, could reveal deeper patterns and interactions. Furthermore, comparing the SHAP method with other interpretability techniques, such as LIME or Integrated Gradients, would offer a more comprehensive picture of the advantages and limitations of each methodology. Finally, integrating the present approach into automated design tools could significantly accelerate optimization processes, making artificial intelligence an integral part of analog integrated circuit design.

This work demonstrates that the intersection of machine learning with explainable artificial intelligence opens new perspectives in the field of microelectronics. Its contribution lies not only in increasing efficiency but also in enhancing understanding, an element necessary for the evolution of innovative and reliable technologies.

Chapter 6

Bibliography

Bibliography

- [6.1] J. Wei, Y. Zeng, and X. Liu, “An enhanced operational amplifier macro model based on artificial neural network,” *International Journal of Circuit Theory and Applications*, 50(12): 5742–5755, 2022. DOI: <https://doi.org/10.1002/cta.3381>.
- [6.2] J. P. da Silva Rosa, *Using Artificial Neural Networks to Size Analog Integrated Circuits*, M.Sc. Dissertation, Instituto Superior Técnico, Universidade de Lisboa, 2018. Available: https://fenix.tecnico.ulisboa.pt/downloadFile/844820067125542/Dissertation_74149.pdf.
- [6.3] S. Rakhimov, M. Shahid, and A. Riaz, “Machine Learning Driven Global Optimisation Framework for Analog Circuit Design,” *arXiv:2404.02911*, 2024. Available: <https://arxiv.org/abs/2404.02911>.
- [6.4] A. Wang and X. Tang (eds.), “Using Artificial Neural Networks for Analog Integrated Circuits Design,” in *Automatic Sizing and Layout of Analog ICs using Deep Learning*, Springer, 2020. Available (chapter PDF): <https://link.springer.com/content/pdf/10.1007/978-3-030-35743-6.pdf>.
- [6.5] A. Esteva, A. Robicquet, B. Ramsundar, V. Kuleshov, M. DePristo, K. Chou, C. Cui, G. Corrado, S. Thrun, and J. Dean, “A guide to deep learning in healthcare,” *Nature Medicine*, 25(1):24–29, 2019. DOI: <https://doi.org/10.1038/s41591-018-0316-z>. Available: <https://www.nature.com/articles/s41591-018-0316-z>.
- [6.6] S. Grigorescu, B. Trasnea, T. Cocias, and G. Macesanu, “A survey of deep learning techniques for autonomous driving,” *Journal of Field Robotics*, 37(3):362–386, 2020. DOI: <https://doi.org/10.1002/rob.21918>. Available: <https://onlinelibrary.wiley.com/doi/abs/10.1002/rob.21918>.
- [6.7] T. Fischer and C. Krauss, “Deep learning with long short-term memory networks for financial market predictions,” *European Journal of Operational Research*, 270(2):654–669, 2018. DOI: <https://doi.org/10.1016/j.ejor.2017.11.054>. Available: <https://www.sciencedirect.com/science/article/pii/S0377221717310652>.
- [6.8] M. T. Ribeiro, S. Singh, and C. Guestrin, “Why Should I Trust You? Explaining the Predictions of Any Classifier,” in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pp. 1135–1144, 2016. DOI: <https://doi.org/10.1145/2939672.2939778>. Available: <https://dl.acm.org/doi/10.1145/2939672.2939778>.
- [6.9] M. Sundararajan, A. Taly, and Q. Yan, “Axiomatic Attribution for Deep Networks,” in *International Conference on Machine Learning (ICML)*, PMLR, pp. 3319–3328, 2017. Available: <https://proceedings.mlr.press/v70/sundararajan17a.html> (PDF: <https://proceedings.mlr.press/v70/sundararajan17a/sundararajan17a.pdf>).
- [6.10] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, “Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization,” in *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pp. 618–626, 2017. Available (OpenAccess PDF): https://openaccess.thecvf.com/content_ICCV_2017/papers/Selvaraju_Grad-CAM_Visual_Explanations_ICCV_2017_paper.pdf.

- [6.11] W. McKinney, “Data Structures for Statistical Computing in Python,” in *Proceedings of the 9th Python in Science Conference (SciPy)*, S. van der Walt and J. Millman (eds.), pp. 51–56, 2010. Available: <https://conference.scipy.org/proceedings/scipy2010/mckinney.html> (PDF mirror: <https://proceedings.scipy.org/articles/proceedings-2010.pdf>).
- [6.12] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimelshein, L. Antiga, *et al.*, “PyTorch: An Imperative Style, High-Performance Deep Learning Library,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 32:8024–8035, 2019. Available: <https://papers.nips.cc/paper/9015-pytorch-an-imperative-style-high-performance-deep-learning-library> (PDF: <https://papers.neurips.cc/paper/9015-pytorch-an-imperative-style-high-performance-deep-learning-library.pdf>).
- [6.13] D. P. Kingma and J. Ba, “Adam: A Method for Stochastic Optimization,” in *International Conference on Learning Representations (ICLR)*, 2015. Available: <https://arxiv.org/abs/1412.6980>.
- [6.14] IBM, “Neural Networks — IBM Think,” 2023. Available: <https://www.ibm.com/think/topics/neural-networks> (Accessed: Aug. 27, 2025).
- [6.15] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. MIT Press, 2016. Available: <https://www.deeplearningbook.org/>.
- [6.16] L. W. Nagel, “SPICE2: A Computer Program to Simulate Semiconductor Circuits,” UCB/ERL Memorandum M520, University of California, Berkeley, 1975. Available: <https://www2.eecs.berkeley.edu/Pubs/TechRpts/1975/ERL-m-520.pdf>.
- [6.17] W. Liu, X. Jin, K. M. Cao, and C. Hu, “BSIM4.1.0 MOSFET Model — User’s Manual,” UCB/ERL M00/48, University of California, Berkeley, Oct. 2000. Available: <https://www2.eecs.berkeley.edu/Pubs/TechRpts/2000/ERL-00-48.pdf>.
- [6.18] L. S. Shapley, “A Value for n-Person Games,” in *Contributions to the Theory of Games II* (Annals of Mathematics Studies, No. 28), Princeton University Press, 1953, pp. 307–317.
- [6.19] S. M. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017. Available: <https://arxiv.org/abs/1705.07874>.
- [6.20] S. M. Lundberg and S.-I. Lee, “A Unified Approach to Interpreting Model Predictions,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017. Available: <https://arxiv.org/abs/1705.07874> (PDF: <https://proceedings.neurips.cc/paper/7062-a-unified-approach-to-interpreting-model-predictions.pdf>).
- [6.21] I. Covert and S.-I. Lee, “Improving KernelSHAP: Practical Shapley Value Estimation Using Linear Regression,” in *Proceedings of the 24th International Conference on Artificial Intelligence and Statistics (AISTATS)*, vol. 130, pp. 3457–3465, 2021. Available: <https://arxiv.org/abs/2012.01536> (PDF: <https://proceedings.mlr.press/v130/covert21a/covert21a.pdf>).
- [6.22] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, “Scikit-learn: Machine Learning in Python,” *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011. Available: <https://jmlr.org/papers/volume12/pedregosa11a/pedregosa11a.pdf>

List of Figures

2.1	Architecture of a typical neural network [14].	9
3.1	Operational Amplifier Circuit.	14
3.2	Convergence curve of the cost function (MSE) during the training of the neural network for 3000 epochs.	17
4.1	Comparison of actual and predicted performance metric values on the test set.	21
4.2	SHAP summary plot for the model output.	22
4.3	Phase Margin	24
4.4	Unity Gain Bandwidth	25
4.5	Gain	26
4.6	ICMR max	27
4.7	ICMR min	28
4.8	Vout max	29
4.9	Vout min	30
4.10	Slew Rate	30